

## What is this?

*This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to [stefano.covino@inaf.it](mailto:stefano.covino@inaf.it).*

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# TIME-DOMAIN ASTROPHYSICS

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# Time-Series

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- A time-series is any sequence of observation such that the distribution of a given value depends on the previous values.
- Time is an exogenous (outside the model) variable that is directional - measurements only depend on the past.
  - This is a statement of causality.
- However, the exogenous variable can be anything.
- Let's assume to have a set of data extracted from  $y(t) = A \sin(\omega t)$  with homoscedastic variance  $V = \sigma^2 + A^2/2$ .
  - This is easy to prove if you compute the variance as  $\sum(y - \langle y \rangle)^2/N$ . Since the average value is zero, this turns out to be  $V = \frac{A^2}{N} \sum \sin^2(\omega t)$  giving the  $A^2/2$  term.
- We can compute the  $\chi^2$  for this toy model:

$$\chi^2_{\text{dof}} = \frac{1}{N} \sum (y/\sigma)^2 = \frac{1}{N\sigma^2} \sum (y)^2 = \frac{1}{\sigma^2} VAR = 1 + \frac{A^2}{2\sigma^2}$$

- With no variability ( $A \sim 0$ ) the expectation value of the  $\chi^2_{\text{dof}} \sim 1$  with standard deviation  $\sqrt{2/N}$ , while it'll be larger in case of variability.
- Therefore, in order to have  $\chi^2_{\text{dof}} > 1 + 3\sqrt{2/N}$  we need  $A > \sigma\sqrt{72/N}$ , which shows that with  $N$  sufficiently large we can detect variability well below the uncertainty of the single points.

# Fourier Analysis

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- The aim of Fourier analysis is to express any function as a sum of different sines and cosines, characterised by an angular frequency  $\omega$  or a corresponding time period  $P = 2\pi/\omega$ .
- Such a decomposition is known as a Fourier series:

$$f(t) = \frac{a(0)}{2} + \sum_{n=1}^{\infty} [a(n) \sin n\omega_0 t + b(n) \cos n\omega_0 t]$$

- This expression can represent practically any periodic function with period  $P_0$ , with suitable adjustment of the coefficients  $a(n)$  and  $b(n)$ , which are known as the Fourier coefficients.
- The conditions under which the Fourier decomposition is valid are that  $f(t)$  has only a finite number of finite discontinuities and only a finite number of extreme values within a period.
  - These are known as Dirichlet conditions and the functions obeying them are called piece-wise regular.
- The Fourier coefficients  $a(n)$  and  $b(n)$  can be determined by performing the following integrations:

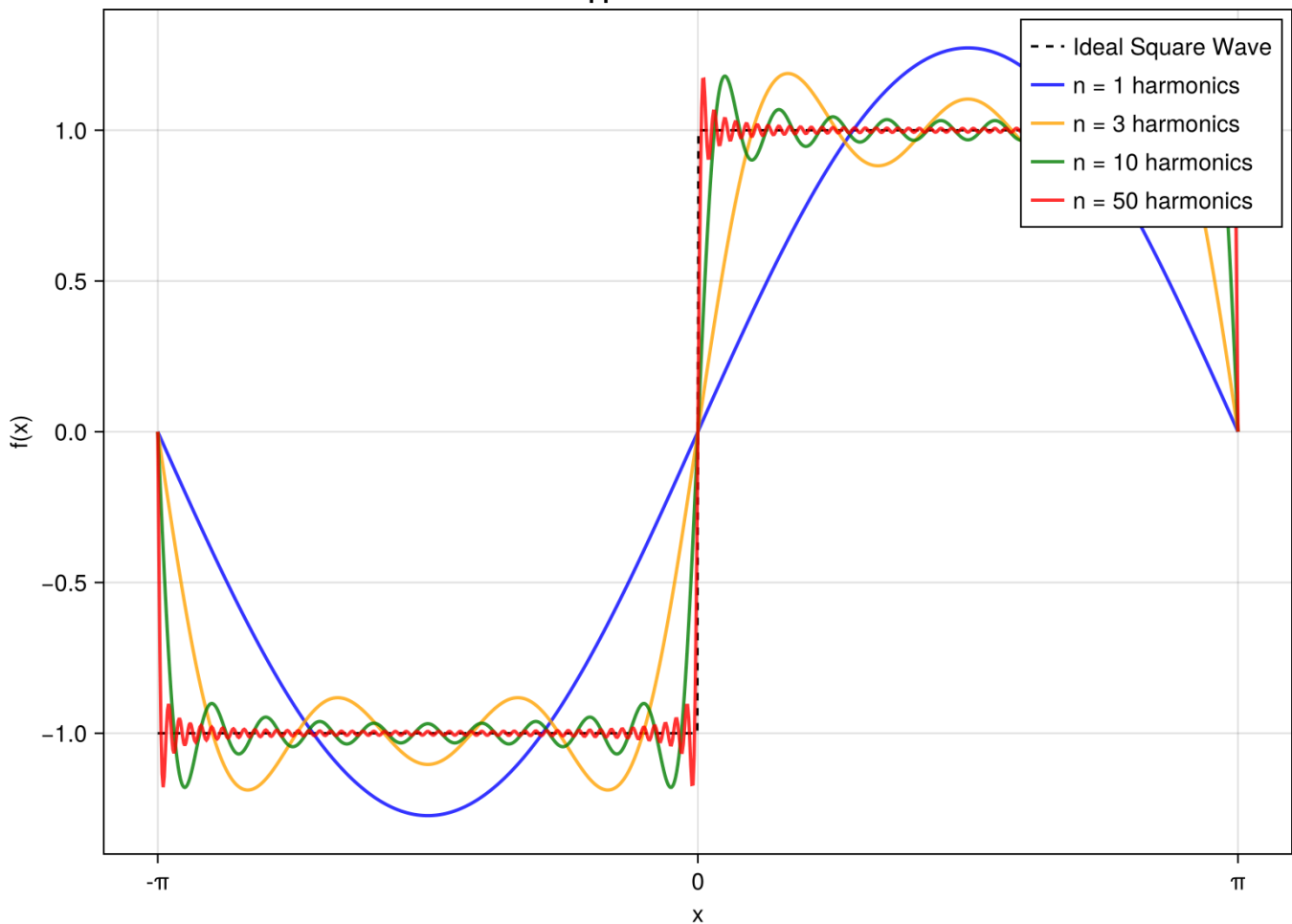
$$a(n) = \frac{2}{P_0} \int_0^{P_0} f(t) \sin n\omega_0 t dt ; \quad b(n) = \frac{2}{P_0} \int_0^{P_0} f(t) \cos n\omega_0 t dt, \quad n = 0, 1, 2, \dots$$

- Alternatively, the Fourier decomposition may also be expressed in an equivalent exponential form:

$$f(t) = \frac{1}{P_0} \sum_{n=-\infty}^{\infty} c(n) e^{in\omega_0 t}; \quad \text{where } c(n) = \int_0^{P_0} f(t) e^{-in\omega_0 t} dt$$

- In the above expansions, the  $n = 0$  term is often called the constant or the D.C. (Direct Current) component, the  $n = 1$  term the fundamental and the terms with  $n > 1$  the harmonics.

Fourier Series Approximation of a Box-Car Function



- Example of a Fourier series decomposition. The original time domain function  $f(t)$ , shown in the dashed line, is a square wave, which equals 1 from  $t = 0$  to  $\pi$  and -1 from  $-\pi$  to 0. Thin lines show the Fourier sum with different number of terms.
  - The gradual improvement in the approximation of the function with increasing number of terms in Fourier series is evident.
- At the point of discontinuity, all the reconstructions pass through the average of the left and right limits of the original function and the transition gets progressively sharper with larger number of terms.

## Continuous Fourier Transform

- We define the Fourier transform (FT) of any function  $f(t)$  as:

$$F(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t} dt$$

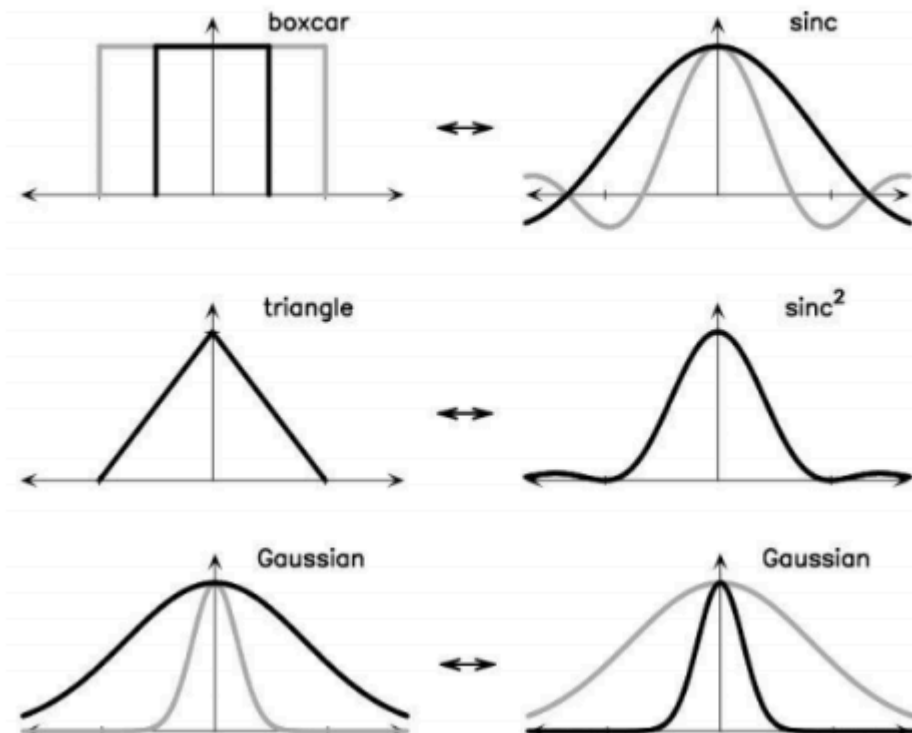
- This is a linear transformation and no information is lost. The representations of a function in time and frequency domains are equivalent.
- The original  $f(t)$  can be recovered by applying the inverse Fourier transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega)e^{i\omega t} d\omega$$

- The FT has a number of interesting properties. It is linear, not necessarily a real function, and its amplitude is invariant to time shift (but not its phase).

| $f(t)$        | $F(\omega)$                |
|---------------|----------------------------|
| Real          | $H(-\omega) = H^*(\omega)$ |
| Even          | Even                       |
| Odd           | Odd                        |
| Real and Even | Real and Even              |
| Real and Odd  | Imaginary and Odd          |

- Unless the original function is even, an unlikely situation in the case of a time series, its Fourier transform is complex!



- Scaling:  $h(at) \longleftrightarrow H(f/a)/|a|$ , "broad"  $\longleftrightarrow$  "narrow"
- Shifting:  $h(t - t_0) \longleftrightarrow H(f) * e^{2\pi i f t_0}$ , "shift"  $\longleftrightarrow$  "phase roll/gradient"

- Convolution:  $h(t) * g(t) \longleftrightarrow H(f)G(f)$ , "convolution"  $\longleftrightarrow$  "multiplication"

How to compute the Fourier Transform in a few simple cases can be found here ([notebook, html](#)).

## Power density spectrum (PDS)

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- The Power Density Spectrum (PSD) is defined as the Fourier Transform multiplied by its complex conjugate and therefore the square modulus of the Fourier Transform:

$$P(\omega) = F(\omega) \cdot F^*(\omega) = |F(\omega)|^2$$

- If the original function is real (usually the case for time series) the PSD is an even function and the values at negative frequencies are redundant.
- The FT is a linear function, the PDS is not. This means that while the FT of the sum of two signals is the sum of the FT of the signals, in the case of the PDS this is not true and there are cross-terms to be considered.
- E.g., if two signals are  $f(t)$  and  $g(t)$ , the PSD of the sum of the two is:

$$\begin{aligned} P[f(t) + g(t)] &= |F[f(t) + g(t)]|^2 = F[f(t) + g(t)] \cdot F^*[f(t) + g(t)] = \\ &= P[f(t)] + P[g(t)] + 2\text{Re}\{F[f(t)] \cdot F^*[g(t)]\} \end{aligned}$$

- If the two signals are uncorrelated, the cross-term is zero and linearity applies.

## Autocorrelation function (ACF)

---

- The autocorrelation (ACF) of a function  $f(t)$  is defined as:

$$A(t) = \int_{-\infty}^{+\infty} f(\tau) f(t + \tau) d\tau \iff F(f) F^*(f) \equiv |F(f)|^2$$

- The autocorrelation of a function is the Fourier transform of its PSD.
- It is also simple to derive, by the *Parseval's theorem*, simply setting  $t = 0$ , that:

$$\int_{-\infty}^{+\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |f(\omega)|^2 d\omega$$

For the interested readers, a proof that PDS and ACF are Fourier duals can be found here ([notebook](#), [html](#)).

## Discrete Fourier transform

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- In the real world, we only have discrete measurements, the time series, commonly called, in astronomy, “light curves”. They consist of  $N$  measurements  $x_k$  taken (now) at equally-spaced times  $t_k$  from  $0$  to  $T$ .
- In this case we can define the discrete Fourier transform (and its inverse) as:

$$a_j = \sum_{k=0}^{N-1} x_k e^{-2\pi i j k / N} \quad (j = -N/2, \dots, N/2 - 1)$$

$$x_k = \frac{1}{N} \sum_{j=-N/2}^{N/2-1} a_j e^{2\pi i j k / N} \quad (k = 0, \dots, N - 1)$$

- Since the data are equally spaced, the times are  $kT/N$  and the frequencies are  $j/T$ .
- The time step is  $\delta t = T/N$  and the frequency step is  $\delta \nu = 1/T$ .
- As the discrete time series has a time step  $\delta t$  and a duration  $T$ , there are limitations to the frequencies that can be examined:
  - The lowest frequency is  $1/T$ , corresponding to a sinusoid with a period equal to the signal duration.
  - The highest frequency that can be sampled, is called *Nyquist frequency*:  $\nu_{\text{Nyq}} = \frac{1}{2\delta t} = \frac{1}{2} \frac{N}{T}$ .
- At the zero frequency, the FT value is just the sum of the signal values:

$$a_0 = \sum_{k=0}^{N-1} x_k e^{-2\pi i 0 k / N} = \sum_{k=0}^{N-1} x_k$$

- *Parseval's theorem* applies also to the discrete case and one can see that the variance of a signal is  $1/N$  times the sum of the  $a_j$  over all indices besides zero (also known as *Plancherel Theorem*):

$$\begin{aligned} \text{Var}(x_k) &= \sum_k (x_k - \bar{x})^2 = \sum_k x_k^2 + \sum_k \bar{x}^2 - \sum_k 2\bar{x}x_k = \sum_k x_k^2 + N\bar{x}^2 - 2N\bar{x}^2 = \\ &= \sum_k x_k^2 - N\bar{x}^2 = \sum_k x_k^2 - \frac{1}{N} \left( \sum_k x_k \right)^2 = \frac{1}{N} \sum_j |a_j|^2 - \frac{1}{N} a_0^2 = \end{aligned}$$

$$\implies \text{Var}(x_k) = \frac{1}{N} \sum_{j=-\frac{N}{2}}^{j=\frac{N}{2}-1} |a_j|^2, j \neq 0$$

**Exercise:** let's compute the Fourier frequency grid for a simple case

- We assume to monitor a given phenomenon observing a quantity of interest for 100s and sampling it every 2s, i.e. we have 50 observations.
- Therefore, the lowest frequency (meaningful to analyse) turns out to be  $1/T = 0.01\text{Hz}$ , and the Nyquist frequency is  $\nu_{\text{Nyq}} = \frac{1}{2\delta T} = \frac{1}{2 \times 2s} = 0.25\text{ Hz}$ .
- The whole ( meaningful) Fourier frequency grid consists therefore of 50 entries, from  $-\nu_{\text{Nyq}}$  to  $+\nu_{\text{Nyq}}$ , excluding the former or the latter value.
- Since for a real input function the power spectrum for negative or positive frequencies is identical, we restrict to the positive side.
- And the Fourier grid is finally, starting from 0 with 25 (i.e.  $N/2$ )  $1/2s = 0.01\text{Hz}$  steps, **0.0, 0.01, 0.02 ... 0.25Hz**.

```

1 begin
2     function generate_noisy_signal(
3         duration::Float64 = 1000.0,
4         fs::Float64 = 100.0,
5         signal_freq::Float64 = 0.5,
6         amplitude::Float64 = 1.0,
7         noise_std::Float64 = 2.0
8     )
9         t = 0:1/fs:duration-1/fs
10        signal = amplitude .* sin.(2π * signal_freq .* t)
11        noise = noise_std .* randn(length(t))
12        noisy_signal = signal .+ noise
13
14        return collect(t), signal, noisy_signal
15    end
16
17
18    # Genera la serie temporale
19    t, clean_signal, noisy_signal = generate_noisy_signal()
20 end;

```

```

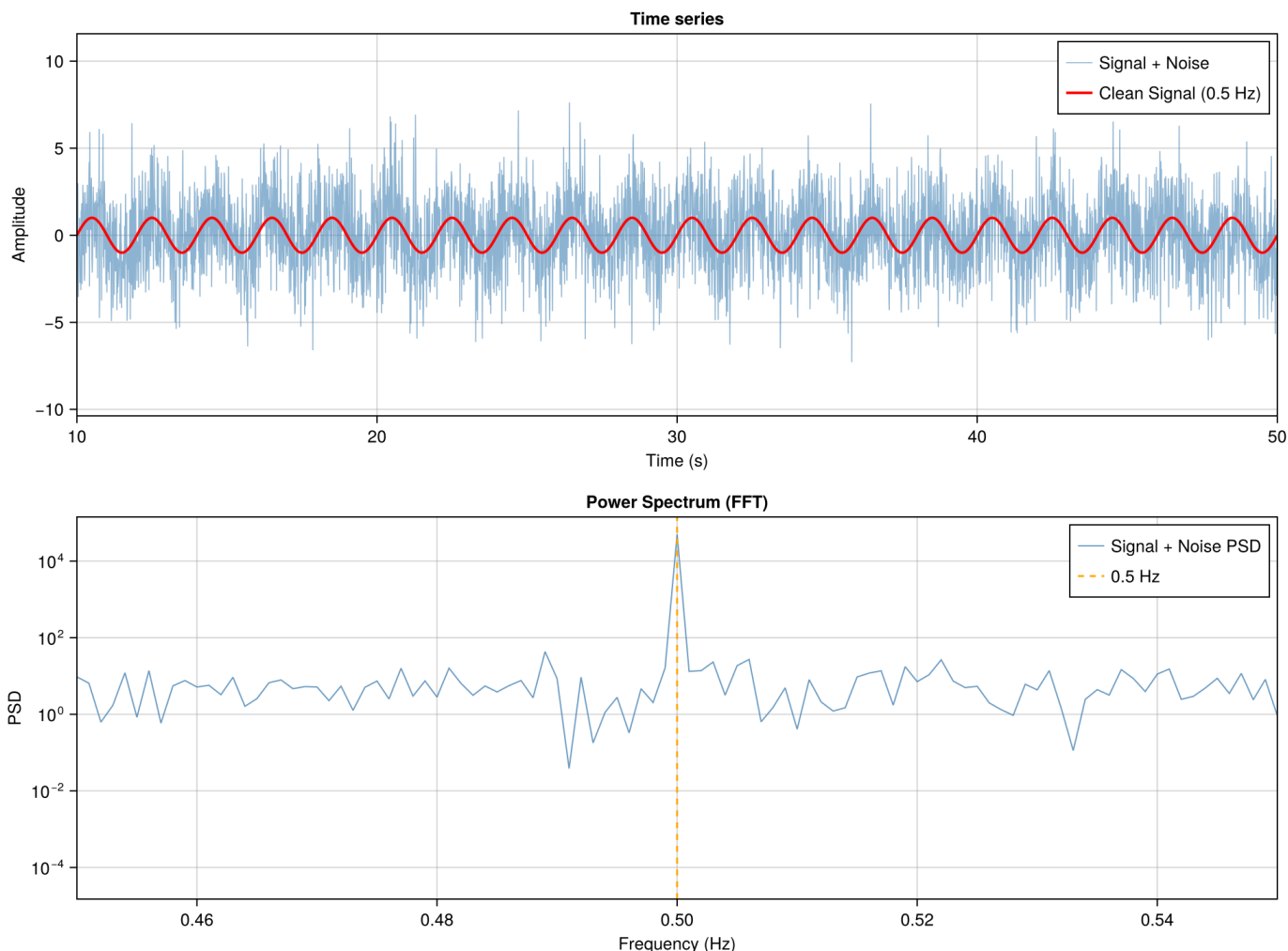
1 function power_spectrum(signal::Vector{Float64}, fs::Float64)
2     N = length(signal)
3
4     # FFT e spettro di potenza (normalizzato)
5     X = fft(signal)
6     psd = (abs.(X).^2) ./ N
7
8     # Solo frequenze positive (metà spettro)
9     freqs = (0:N÷2) .* (fs / N)
10    psd_one_sided = psd[1:N÷2+1]
11
12    # Raddoppia le componenti (eccetto DC e Nyquist) per conservare la potenza
13    psd_one_sided[2:end-1] .*= 2
14
15    return freqs, psd_one_sided
16 end;

```

```

1 begin
2     # --- Calcola spettri ---
3     fs = 100.0
4     freqs_noisy, psd_noisy = power_spectrum(noisy_signal, fs)
5     freqs_clean, psd_clean = power_spectrum(clean_signal, fs)
6 end;

```



```

1 begin
2     fig2 = Figure(size = (1000, 750), fontsize = 13)
3
4     # --- Pannello 1: Serie temporale ---
5     ax1 = Axis(fig2[1, 1],
6         xlabel = "Time (s)",
7         ylabel = "Amplitude",
8         title = "Time series",
9         xgridcolor = (:gray, 0.3),
10        ygridcolor = (:gray, 0.3),
11    )
12
13    lines!(ax1, t, noisy_signal,
14        color = (:steelblue, 0.6), linewidth = 0.8, label = "Signal + Noise")
15    lines!(ax1, t, clean_signal,
16        color = :red, linewidth = 2.0, label = "Clean Signal (0.5 Hz)")
17    axislegend(ax1, position = :rt)
18
19    xlims!(10, 50)
20
21    # --- Pannello 2: Spettro di potenza ---
22    ax2 = Axis(fig2[2, 1],
23        xlabel = "Frequency (Hz)",
24        ylabel = "PSD",
25        title = "Power Spectrum (FFT)",
26        xgridcolor = (:gray, 0.3),
27        ygridcolor = (:gray, 0.3),
28        yscale = log10, # scala logaritmica sull'asse Y
29        xticksvisible = true,
30    )
31
32    lines!(ax2, freqs_noisy, psd_noisy,

```

```

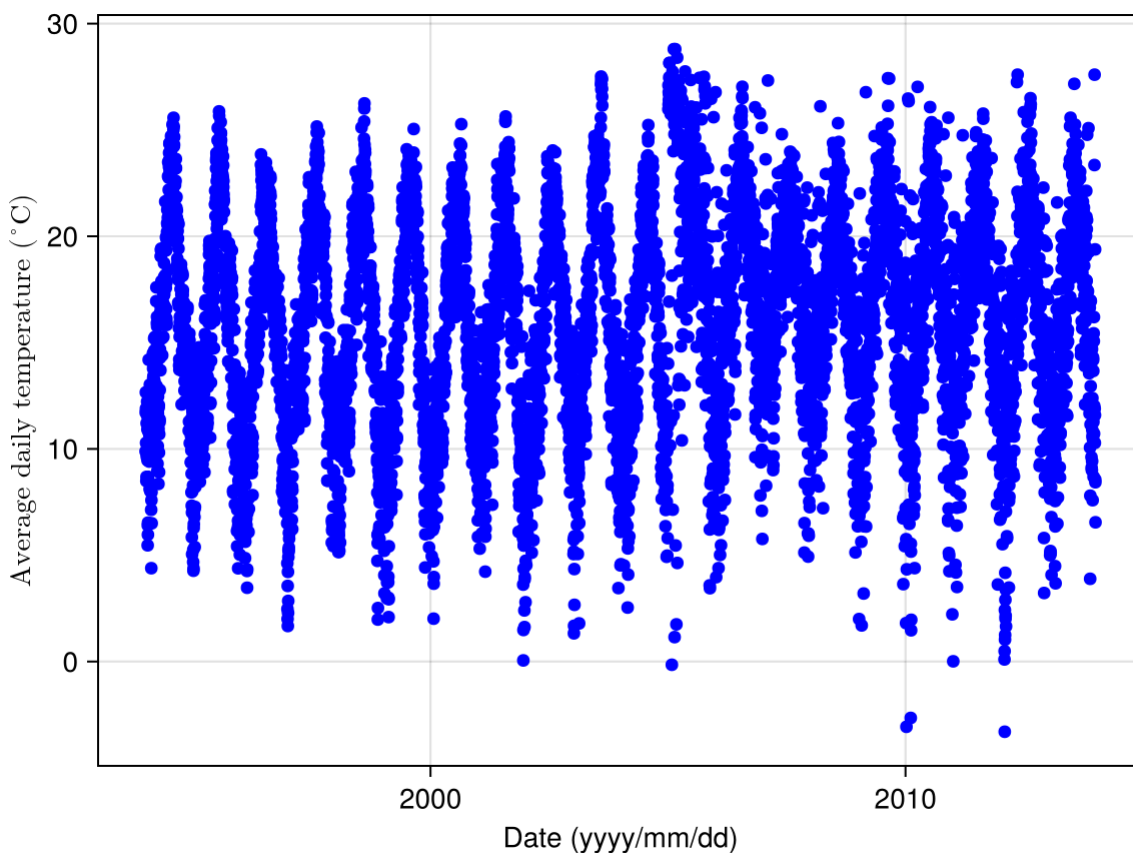
33     color = (:steelblue, 0.8), linewidth = 1.0, label = "Signal + Noise PSD")
34     #lines!(ax2, freqs_clean, psd_clean,
35     #     color = :red, linewidth = 1.5, label = "Clean Signal PSD")
36
37     # Linea verticale sul picco a 0.5 Hz
38     vlines!(ax2, [0.5], color = :orange, linewidth = 1.5, linestyle = :dash, label =
39     "0.5 Hz")
40
41     # Zoom sull'asse X fino a 1 Hz per vedere bene il picco
42     xlims!(ax2, 0.45, 0.55)
43     axislegend(ax2, position = :rt)
44     fig2

```

- Top panel: the black line shows 40 seconds of a simulated time series 1000s long, consisting of a weak sinusoidal modulation (red line) "drowned" into a strong Gaussian noise. Bottom panel: corresponding PDS, zoomed on the relevant frequency range, where the modulation is clearly visible. A weak signal spread in time is collected into a single frequency bin at high significance.

## Exercise about an analysis of weather data in France

- In the following exercise, we are going to analyse weather data spanning about 20 years in France obtained from the US National Climatic Data Center.
- Data are imported <http://www.ncdc.noaa.gov/cdo-web/datasets#GHCND>. The number "-9999" is used for N/A values. And we need to parse dates contained in the DATE column
- Let's now select only dates later than Jan 1994, and compute daily averages dropping the missing data and plot our dataset.
  - The temperature unit is in tenths of a degree, and we get the average value between the minimal and maximal temperature.

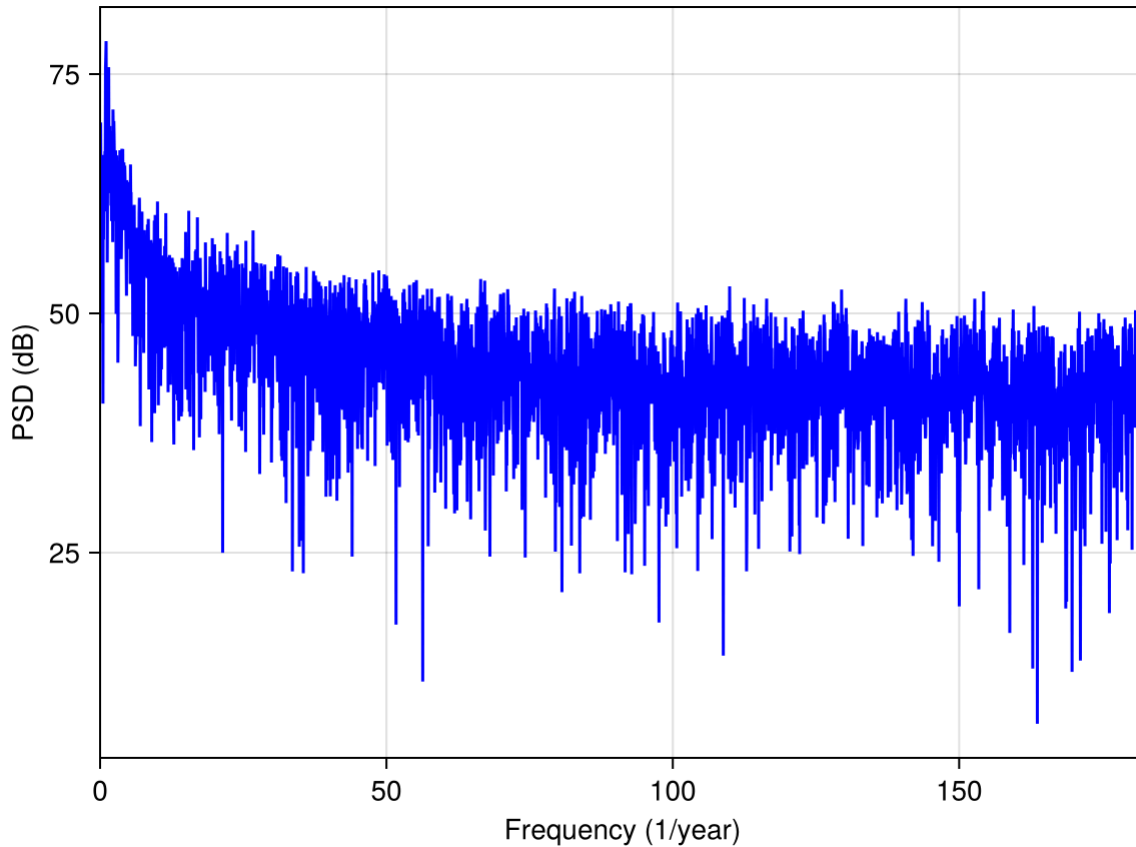


- We now compute the Fourier transform and the spectral density of the signal using the **fft()** function.
- Once the FFT has been obtained, we take the square of its absolute value in order to get the **power spectral density (PSD)**.
- Then, we get the frequencies corresponding to the values of the PSD by the **fftfreq()** utility function.
- Since the original unit is in days we change it to annual unit by the factor **1/365**.
- We restrict to positive frequencies only and let's check which is the maximum frequency for our dataset. Knowing the sampling rate we predict that it should correspond to a period of 2 days so

that the Nyquist frequency turns out to be exactly  $0.5 \text{ day}^{-1}$ .

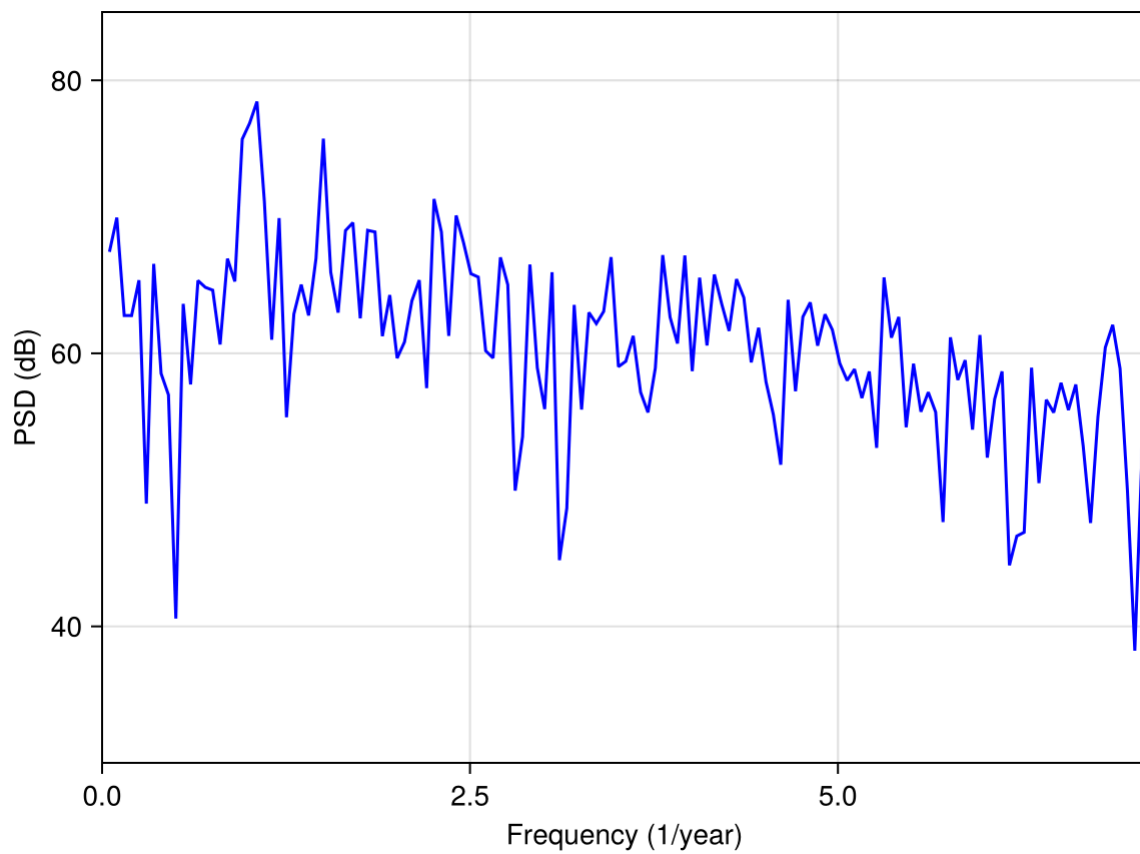
## Nyquist frequency: 0.5

- Let's now plot the power spectral density of our signal, as a function of the frequency (in unit of **1/year**). We choose a logarithmic scale for the y axis (decibels).

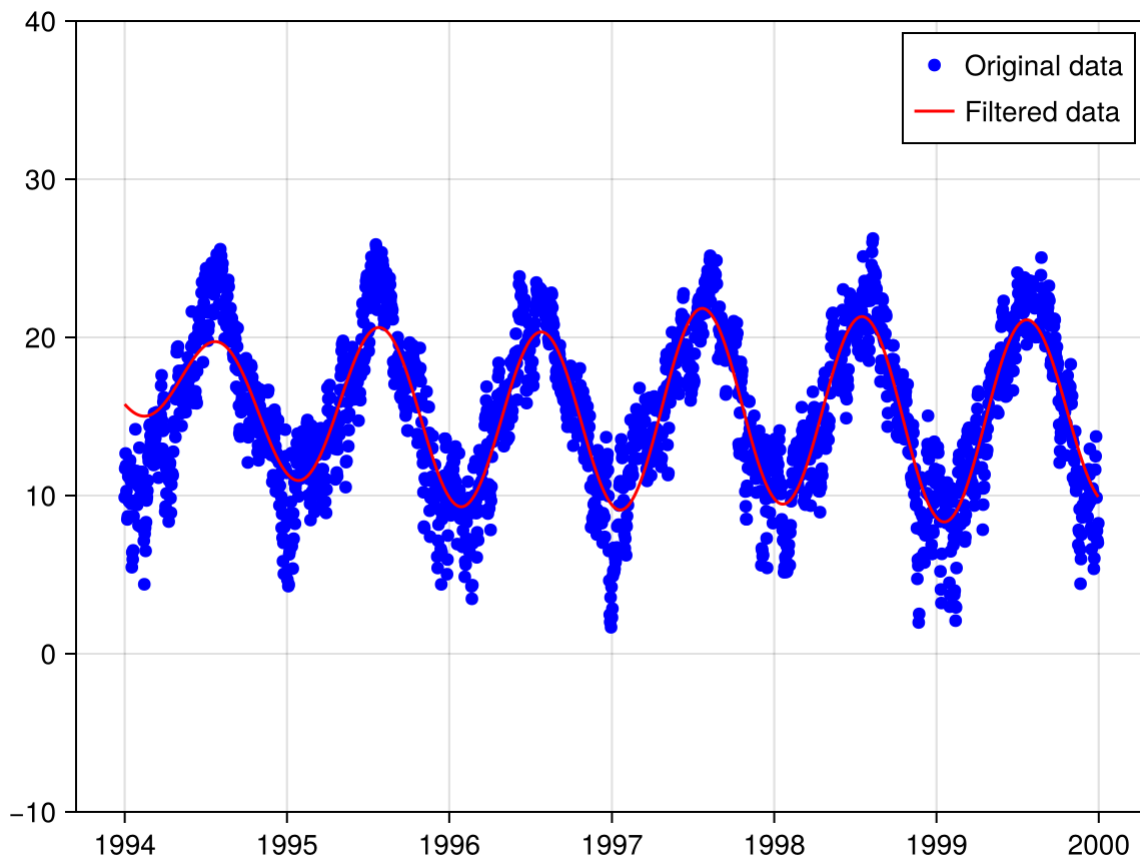


- Or with a better zoom in a region of our interest.

Plot horizontal axis limit:



- Not surprisingly, the fundamental frequency of the signal is the yearly variation of the temperature at  $f=1$ .
- We can now "clean" our data cutting out frequencies higher than the fundamental frequency and by an **inverse FFT** we recover a signal that mainly contains the fundamental frequency.



# What do we observe in the real world?

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# Windowing and Sampling

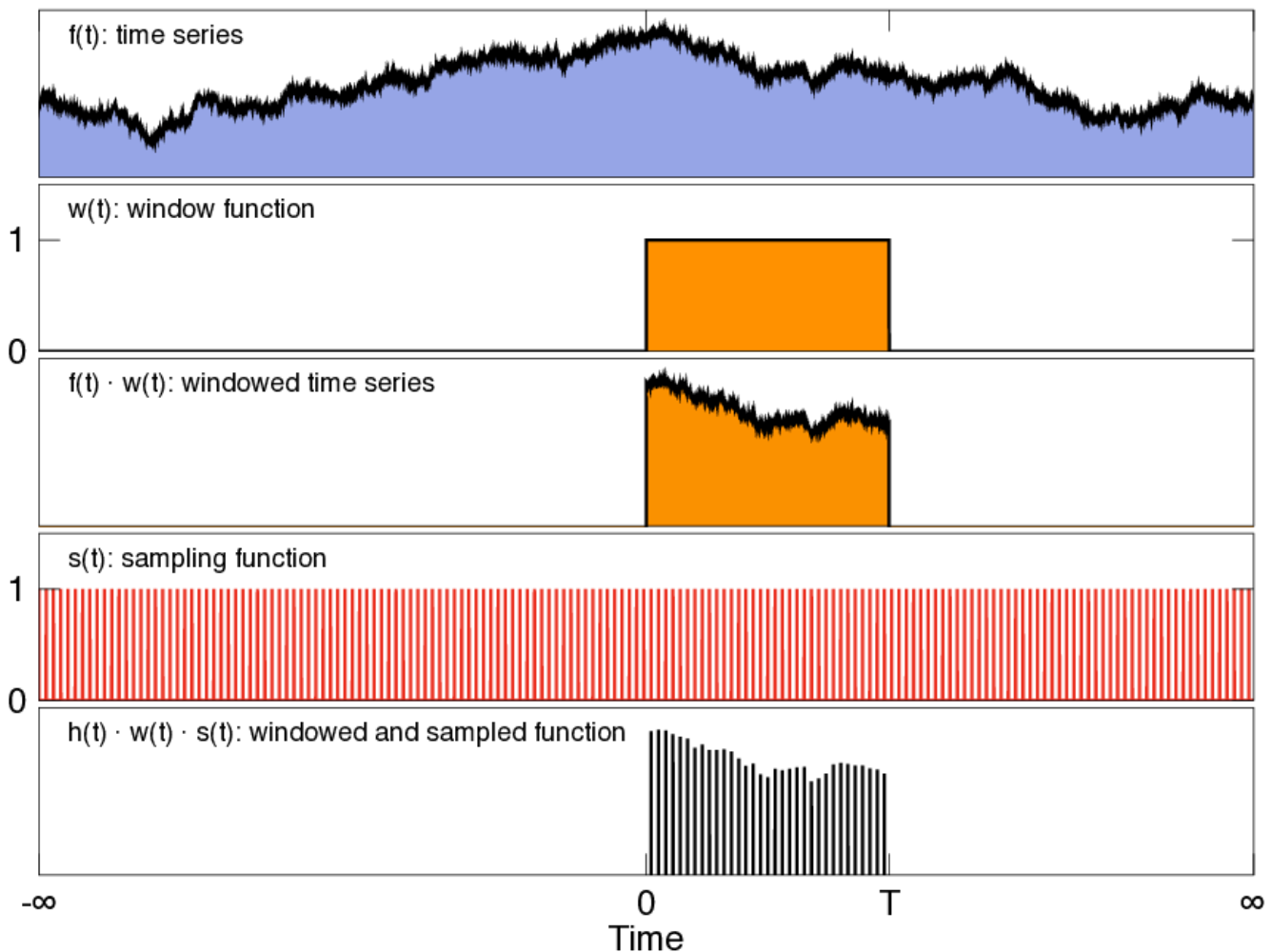
- The CFT and the DFT can be connected easily taking into account that the FT of the product of two functions is the convolution of the FT of the functions.

$$F[x \cdot y] = F[x] \otimes F[y] = \int_{-\infty}^{+\infty} F[x(\nu')] F[y(\nu - \nu')] d\nu'$$

- A discrete time series  $x(t_k) \equiv x_k$  can be seen as the product of a continuous function  $f(t)$  over  $(-\infty, \infty)$  and two additional functions:  $w(t)$  to limit it to the  $(0, T)$  interval and  $s(t)$  to sample it at times  $t_k$ :

$$x_k = h(t) \cdot w(t) \cdot s(t)$$

- $w(t)$  is a boxcar window function, which is 1 in the  $(0, T)$  interval and zero outside.  $s(t)$  is a series of delta functions at  $t_k$ , spaced by  $T/N$ :

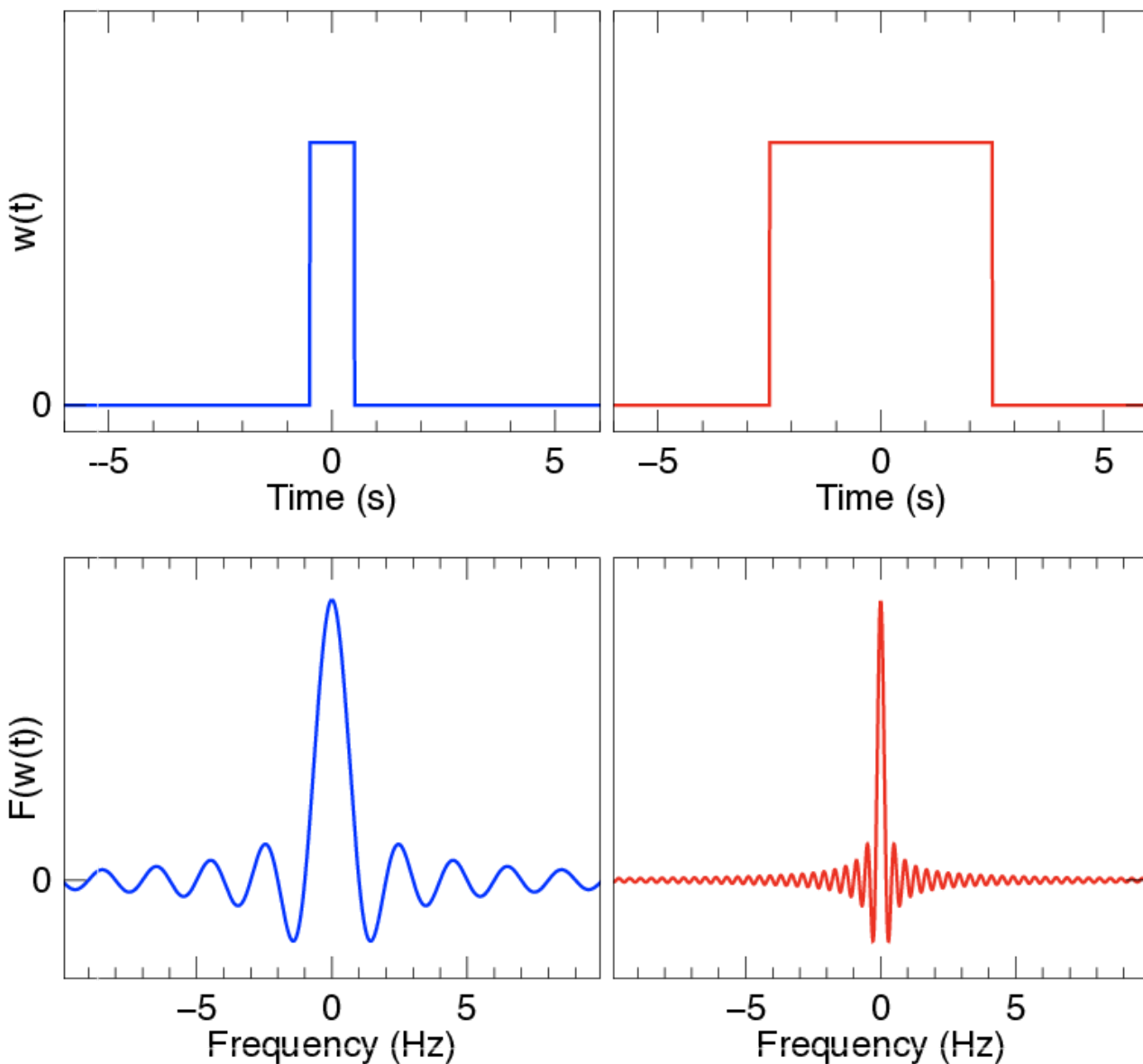


## Windowing effects

- Let us consider a purely sinusoidal function  $f(t) = \sin(\omega t)$ , whose FT is a delta function at  $\omega$ .
- The multiplication by the window function corresponds to the convolution of the delta function with the FT of the window.
- It is simple to calculate the FT of the window: we consider a window function that is unity in the  $-T/2, T/2$  interval, as it is a real and even function, whose FT is also real and even:

$$F(w(t)) = 2 \frac{\sin(\pi \nu T)}{\pi \nu}$$

- This is the well known “sinc” function.
- An important general rule is that the FT peak is broader for shorter T. Something easily deducible from the general properties of FT.
  - The resolution of the signal FT is therefore higher the longer the observation is.
- In addition to the broadening, there is the formation of side lobes. They are much lower than the central peak, but cannot always be ignored.

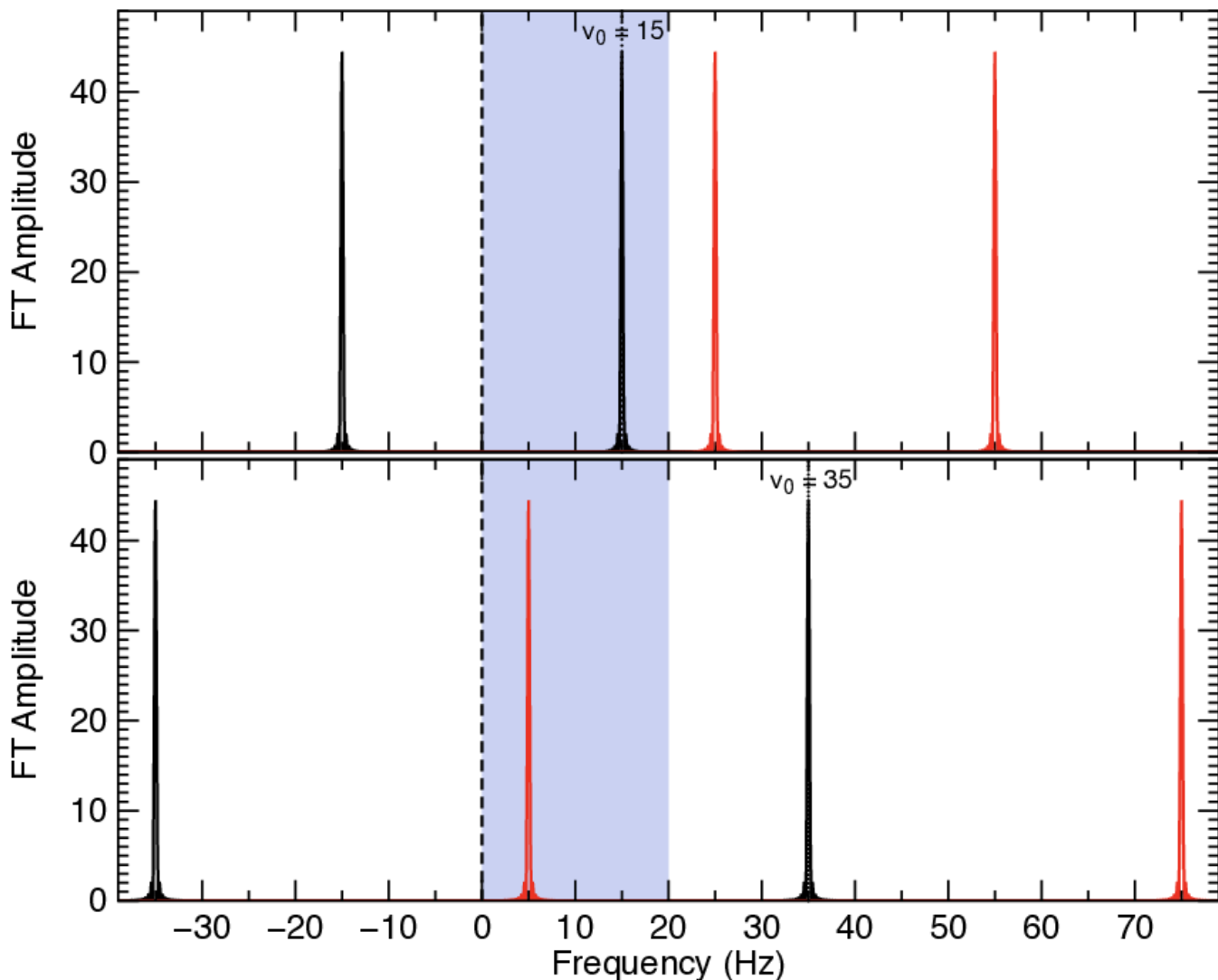


## Sampling effects: aliasing

- The FT of a series of regularly spaced delta functions with spacing  $T/N$  is itself a series of delta functions with spacing  $N/T$ :

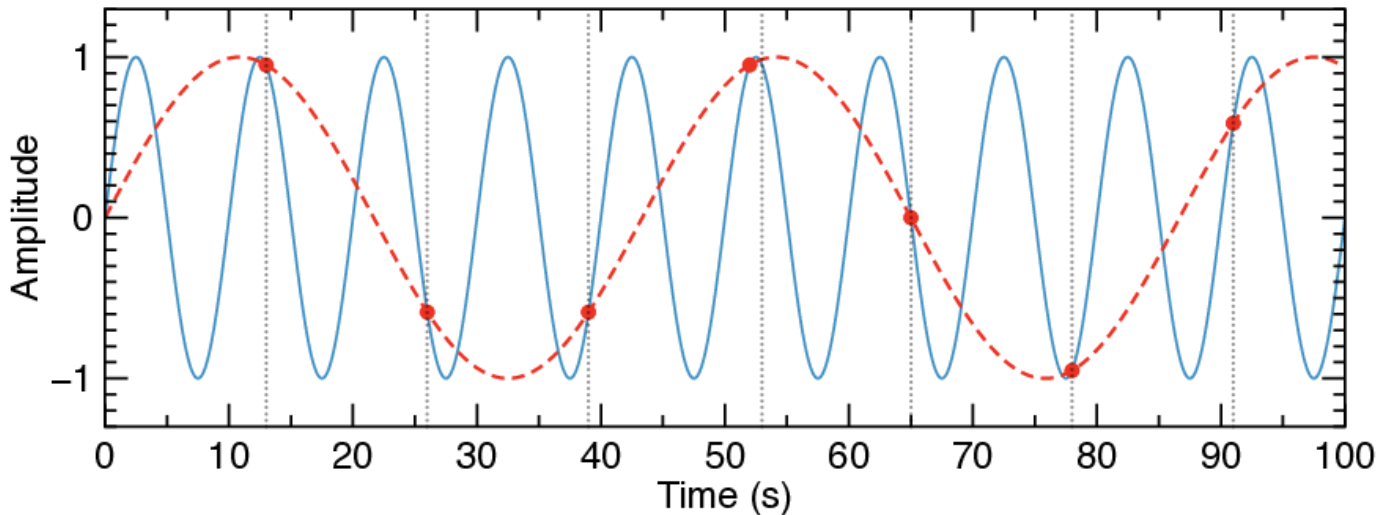
$$s(t) = \sum_{k=-\infty}^{+\infty} \delta(t - \frac{kT}{N}) \iff F(s(t)) = \sum_{m=-\infty}^{+\infty} \delta(\nu - \frac{mN}{T})$$

- Therefore, the effect of sampling on the FT of a sinusoidal signal with frequency  $\nu_0$  (a delta function at  $\nu_0$ ) is that of adding an infinite sequence of delta functions spaced by  $N/T$ , called *aliases*.
- Depending on the frequency of the original signal and the Nyquist frequency we can have different situations.
  - In fact, features at  $\nu = \nu_{N/2} + \nu_x$  also appear at  $\nu = \nu_{N/2} - \nu_x$ .
- This happens because the transition from the CFT to the DFT involves two operations:
  - windowing, a convolution with the function  $W(f)$ , which is essentially a peak with a width  $\delta f = 1/T$  plus sidelines,
  - and aliasing, a reflection of features above the Nyquist frequency back into the range  $(0, \nu_{N/2})$ .



- Top panel: Aliasing on a sinusoidal signal at 15 Hz sampled at 40 Hz. In black the true signal FT amplitude, in red the aliased one. The blue region marks the interval below  $\nu_{Nyq}$ . The signal is detected and the aliases are not.

- Bottom panel: same as the top panel, but with a signal at 35 Hz. Here the signal is not detected, but the 5 Hz alias is.
- We have all experienced aliasing effects when looking at fast rotating objects like an air fan under fluorescent light.
  - The light provides a sampling at 50 Hz (or 60 Hz, depending on where you live), while the fan has a periodicity.
  - Depending on its angular speed, you can see it rotating apparently much slower, or even to stop and rotate in the opposite direction.



- Time domain example of aliasing. The blue signal has  $\nu_0 = 0.1$  Hz. If it is sampled with  $\nu_{Nyq} = 0.038$  Hz (red points) the red dashed alias is the best fit to the data, with  $\nu_a = 0.023$  Hz.
- In the real world we do not really sample signals, but integrate them over finite time bins, i.e. we convolve it with a binning function:

$$b(t) = \begin{cases} \frac{N}{T} & t \in [-\frac{T}{2N}, \frac{T}{2N}] \\ 0 & \text{outside} \end{cases}$$

- Therefore, the signal FT will be multiplied by that of the binning function, which is again a sinc function:

$$B(\nu) = \frac{\sin \pi \nu / 2 \nu_{Nyq}}{\pi \nu / 2 \nu_{Nyq}}$$

• 
$$B(\nu)$$

is a broad function that reaches 0 at  $2\nu_{Nyq}$  and has the value of  $2/\pi$  at  $\nu_{Nyq}$ .

## Effects of binning on the harmonic analysis

- Binning is a common operation carried out for different reasons, e.g., for casting an irregularly sampled time series to a regular grid, for improving the S/N of each point, to reduce the computational burden, etc.

Nevertheless, binning is never a price-free operation.

- Let's assume we have a dataset defined as  $D = \{y_1, y_2, \dots, y_n\}$ , with observations carried out at a regular time step. Its discrete Fourier transform is:

$$Y(\omega) \equiv \sum_{t=1}^n y_t e^{i\omega t}$$

- This is defined for continuous values of  $\omega$  but we know that no loss of data happens if we compute the Fourier transform at the Fourier grid  $\omega_k \equiv 2\pi k/n$ ,  $0 \leq k < n$ .

## Moving average

- Let's suppose our data are replaced by a moving average of past values (i.e. a rebinning):

$$z_t \equiv \sum_{s=0}^{m-1} y_{t-s} \omega_s$$

- where  $\omega_s$  is the weighting coefficient for lag  $s$ .
- Since this can be seen as a convolution of the input data we know that the Fourier transform of  $z$  is the product of the Fourier transform of the original data and of the average function:

$$Z(\omega) = W(\omega)Y(\omega)$$

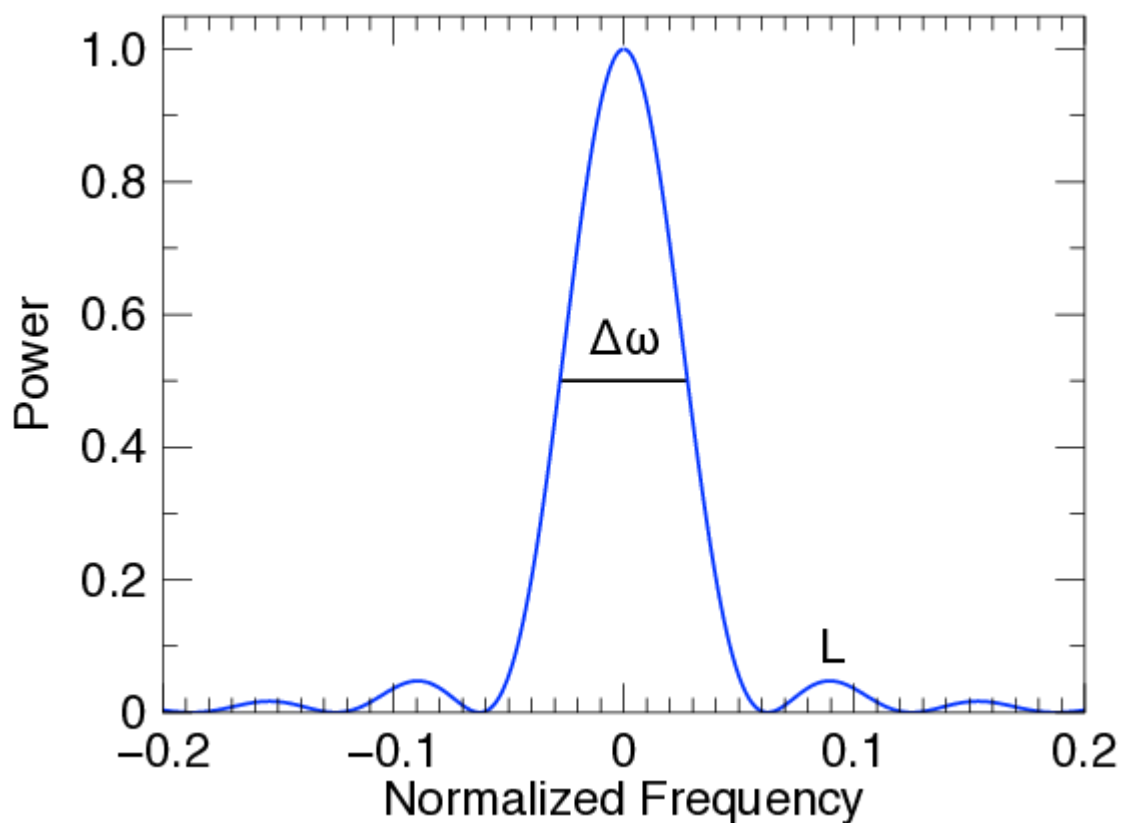
- where  $W(\omega) = \sum_{s=0}^{m-1} \omega_s e^{i\omega s}$ .
- In particular, for uniform weighting,  $\omega_s = 1/m, 0 \leq s < m$ , and we have the well known "sinc" function:

$$W(\omega) = \frac{1}{m} \sum_{s=0}^{m-1} e^{-i\omega s} = e^{-i\frac{\omega}{2}(m-1)} \left[ \frac{\sin(m\omega/2)}{m \sin(\omega/2)} \right]$$

- It is clear, therefore, that together with changing the phase of the Fourier transform, any binning typically decreases the amplitude of the Fourier transform of the original data: i.e. it might be useful for plotting purposes and for saving computer power, yet binning should typically be avoided.

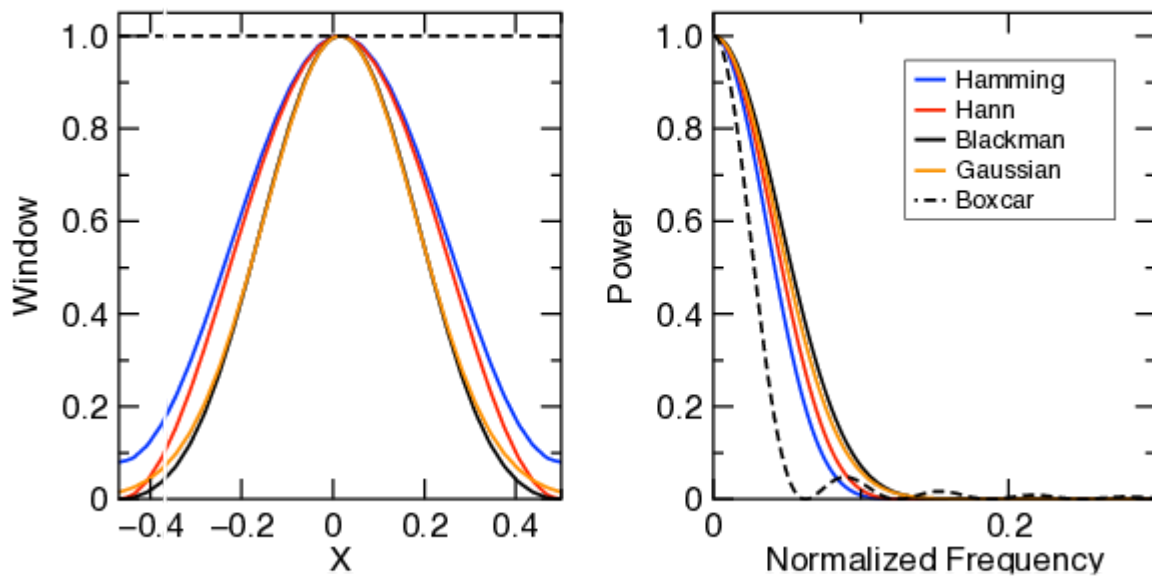
## Window carpentry

- Having a longer observation reduces the width of the main window peak, but does not change the possible spillover effects.
- In some cases it can be advantageous to multiply the data by another window, not boxcar-shaped.
- This results in a loss of signal, as some data are multiplied by a factor less than unity, but there are advantages, depending on the chosen window.
  - Many window functions with different characteristics have been designed and one can tailor them depending on what is needed.
- The main features that identify a window in its PDS are: the width of the main peak  $\Delta\omega$ , the relative amplitude of the first side lobe  $L$  (expressed in decibels) and the slope of the decay of side lobes  $n$ :



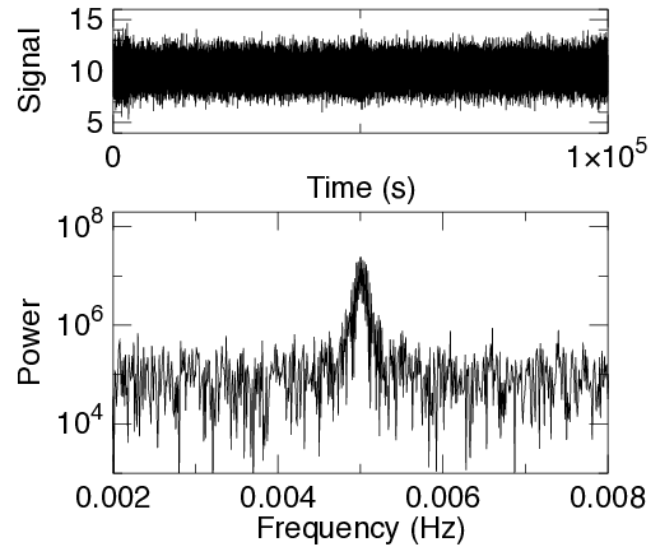
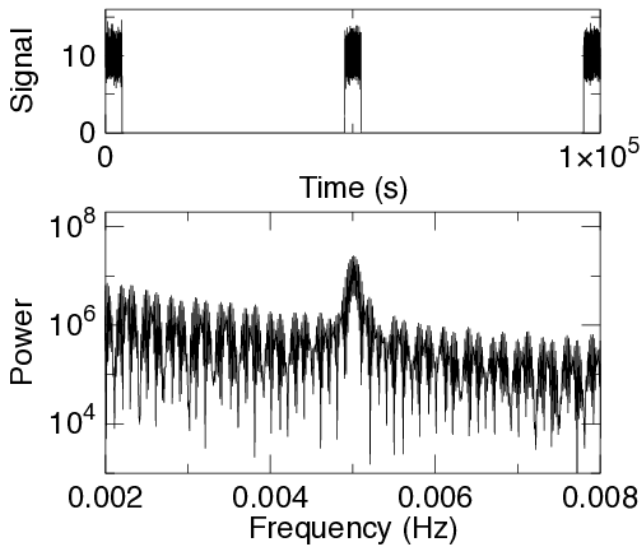
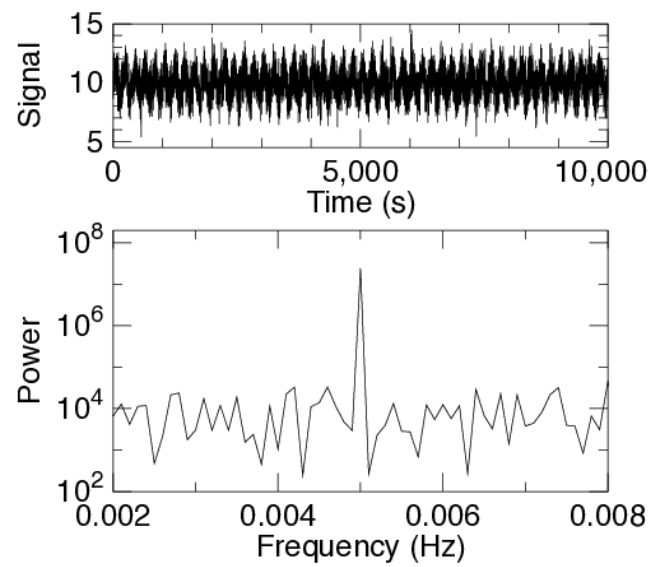
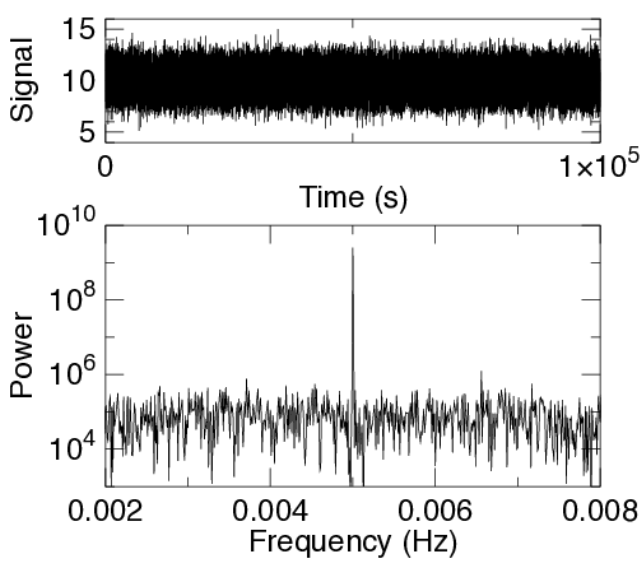
- The boxcar window is the one with the lowest  $\Delta\omega$ , but with alternative windows it is possible to obtain a significant reduction of the amplitude of the side lobes.

| Window   | $\Delta\omega$ | $L$   | $n$ | Function                                |
|----------|----------------|-------|-----|-----------------------------------------|
| Boxcar   | 0.89           | -13db | 2   | 1                                       |
| Hamming  | 1.36           | -43db | 2   | $0.54+0.46 \cos(2\pi t)$                |
| Hann     | 1.44           | -32db | 5   | $0.5(1-\cos(2\pi t))$                   |
| Blackman | 1.68           | -58db | 5   | $0.42+0.5\cos(2\pi t)+0.08\cos(4\pi t)$ |
| Gaussian | 1.55           | -56db | 2   | $\exp(-4.2x^2)$                         |



- Left panel: the shape of five windows: boxcar, Hamming, Hann, Blackman and Gaussian. Right panel: the corresponding PDS, corresponding to (half) the shape of the PDS of a sinusoidal signal. The sidelobes of all but the boxcar window are too low to be seen.

## Effect of gaps



- Each of the four panels contains a signal (top) and its PDS (bottom). In all four, the signal consist of Gaussian noise plus a sinusoid at  $P=200$  s, with 1-s binning.
  - Top left: continuous exposure of  $10^5$  s.
  - Top right: continuous exposure of  $10^4$  s.
  - Bottom left: exposure of  $10^4$  s split into three intervals. The PDS is computed including the gaps as consisting of points at zero level.
  - Bottom right: exposure of  $10^4$  s split into three intervals as in the previous case. The gaps are filled with Gaussian noise.

## Fast Fourier transform

- Evaluation of the Discrete Fourier Transform (DFT) of  $N$  samples involves  $\sim N^2$  multiplication and addition operations – for every Fourier component  $a_j$ , each of the  $N$  samples  $x_k$  needs to be multiplied by a phase factor  $e^{-2\pi i j k / N}$  and then they have to be summed.
- The Fast Fourier Transform algorithm has been devised to accomplish this computation in much fewer steps, typically with  $\sim N \log_2 N$  multiplications and additions. This provides enormous computational savings for large transforms, and has made Fourier analysis accessible to cases where it would have been otherwise prohibitive.

## PSD normalization

- The FT is a linear transformation and the PSD is its squared modulus, the PSD then scales with the square of the intensity level of the signal.
- It is possible to normalize the PSD in different ways. One of the most common is the so-called *Leahy normalization*:

$$P_j^{Leahy} = \frac{2}{N_\gamma} |a_j|^2$$

- where  $N_\gamma$  is the total number of photons in the signal. If instead of counts we have fluxes, and noise is  $\mathcal{N}(0, \sigma)$ ,  $N_\gamma$  is substituted by  $N_{\text{data}} \sigma^2$ .
- The periodogram with this normalization is also known as *Classical* or *Schuster* periodogram.
- With the Leahy normalization the total variance can also be written as:

$$\text{Var}(x_k) = \frac{1}{N} \sum_{j=-\frac{N}{2}}^{\frac{N}{2}-1} |a_j|^2 (j \neq 0) = \frac{N_\gamma}{N} \left( \sum_{j=1}^{\frac{N}{2}-1} P_j + \frac{1}{2} P_{N/2} \right)$$

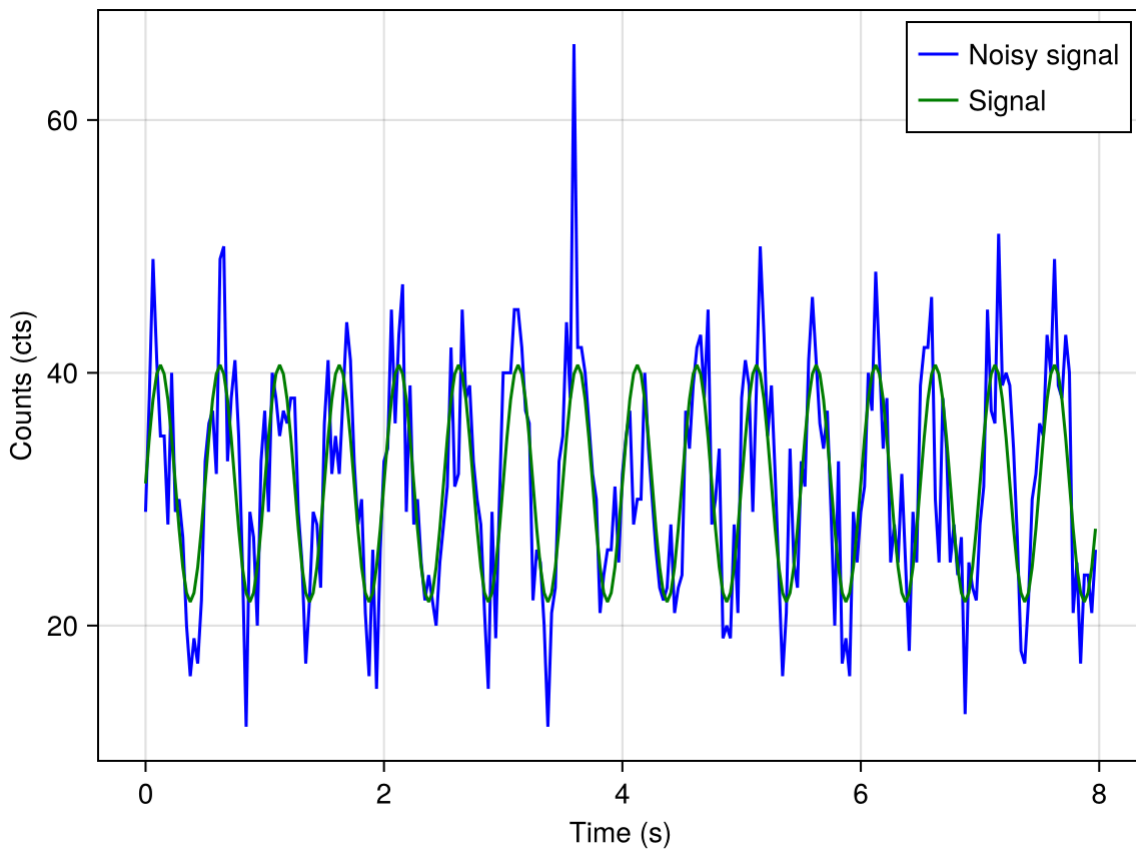
- This normalization leads to a known statistical distribution of signal power:
  - if the signal is dominated by fluctuations due to Poisson statistics and if  $N_\gamma$  is large, powers follow a  $\chi^2$  distribution with 2 degrees of freedom,  $\langle P \rangle = 2$  and  $\text{Var}(P) = 4$ .
- The reason is that the periodogram is the sum of the squares of the real and imaginary parts of the FT and, for a stochastic process, the latter are normally distributed, so the sum of their squares is distributed as a  $\chi^2$  with 2 degrees of freedom.
- For other noise distributions (Poisson, etc.), for large N, due to the central limit theorem the real and imaginary parts become still normal.
- Periodograms are intrinsically very noisy. If the signal is divided into  $S$  segments and the resulting PDS are averaged and rebinned by a factor  $M$ , the powers will be distributed as a  $\chi^2$  with  $2SM$  degrees of freedom scaled by  $1/SM$ : therefore, the average power remains  $\langle P \rangle = 2$ , but the variance is now  $\text{Var}(P) = 4/SM$ .
  - The technique of dividing the time series into equal-duration intervals and averaging the corresponding PSD is called *Bartlett's method*.
- The reduction in time duration  $T$  increases the minimum frequency in the PSD  $\nu_{\min} = 1/T$ .
- This method, in principle, also allows one to skip over (short) data gaps, which have dramatic effects on the PSD.

## Exercise about PSD manipulation

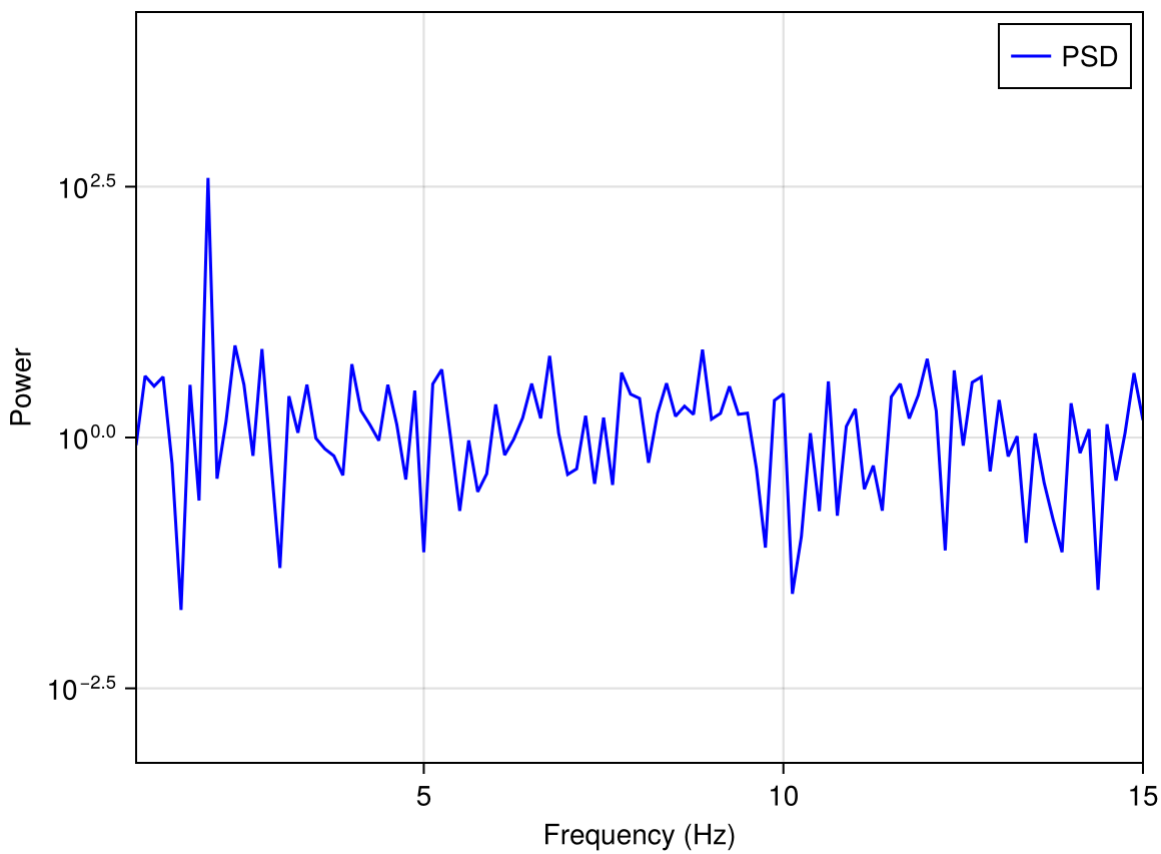
- Let's generate two arrays of relative timestamps, one 8 seconds long and one 1600 seconds long, with  $dt = 0.03125$  s, and make two signals in units of counts. The signal is a sine wave with amplitude = 300 cts/s, frequency = 2 Hz, phase offset = 0 radians, and mean = 1000 cts/s. We then add Poisson noise to the light curve and plot the shortest of the two.

Frequency for the test signal:

Amplitude for the test signal:



- Now let's compute the periodogram

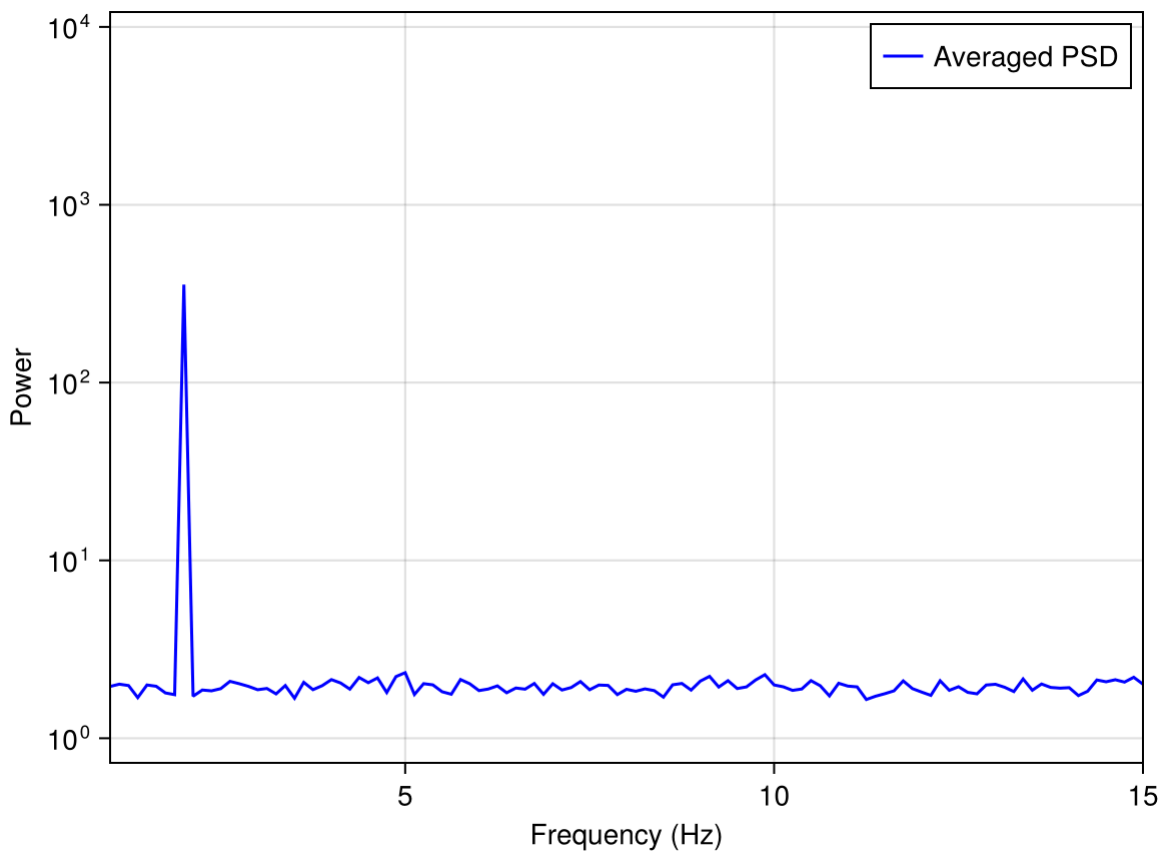


- The computed periodogram (with this function) is normalized so that the area under the periodogram is equal to the uncentered variance (or average power) of the original signal.
- Since the negative Fourier frequencies (and their associated powers) are discarded, the number of time bins per segment  $n$  is twice the length of `freq` and `power`.
- The zero frequency is the sum of the signal value, and it not typically used.

Number of data points: **256**

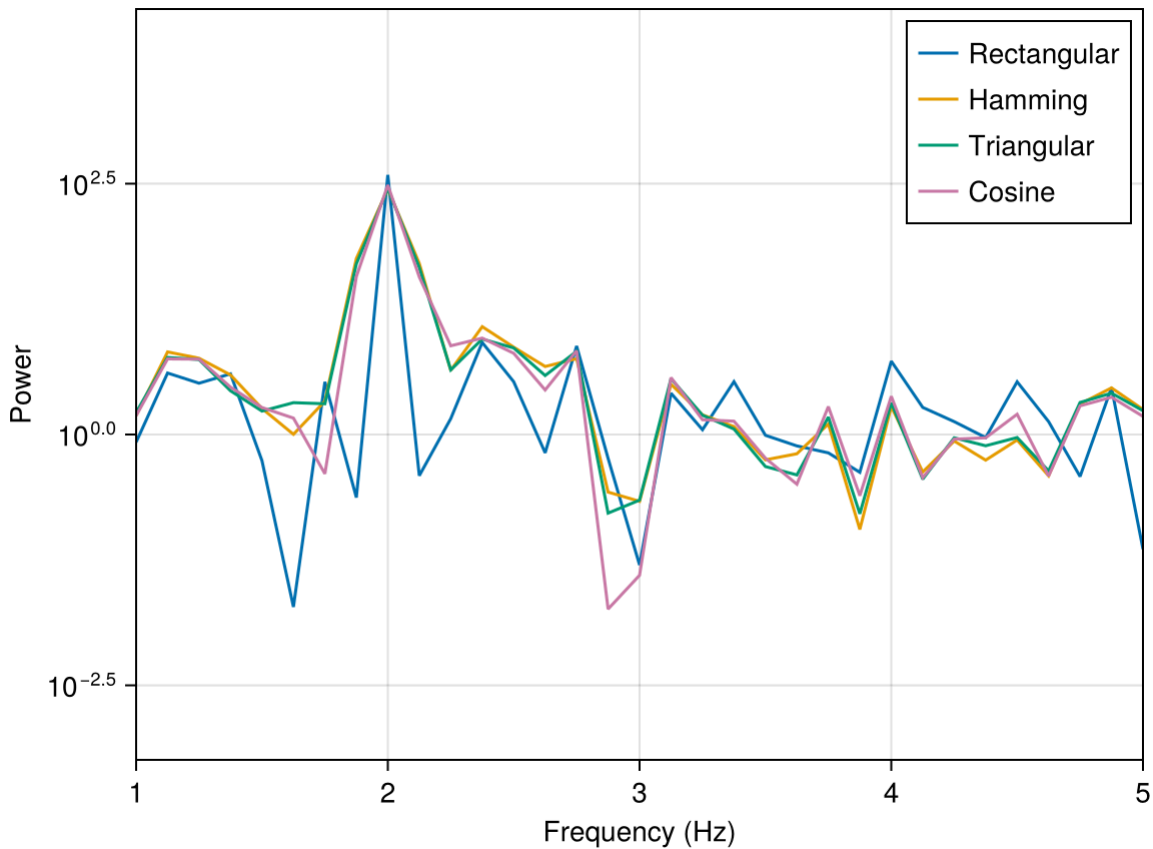
Number of data points: **128**

- The power spectrum is a bit noisy. Let's try averaging together power spectra from multiple segments of data using the long time series.
- We want to average periodograms computed each 8 seconds, averaging therefore  $1600/8 = 200$  periodograms of 256 elements each.



- With a clear increase of the S/N.
- Let's try now to compute periodograms using different windows functions.

2



## Auto and Cross-Correlation

- The Cross-correlation of two functions  $f(t)$  and  $g(t)$  is defined as:

$$C(\tau) = f \star g = \int_{-\infty}^{\infty} f^*(t)g(t + \tau)dt$$

- The result is a function of the *lag*  $\tau$  introduced between the two functions, and is often used to estimate the similarity between two different time series, as a function of lag.
- If a common underlying process causes the time variation of intensity at two different electromagnetic bands with differential delays while propagating to the observer, then the cross correlation function of the two time series will exhibit a peak at the corresponding lag, namely the relative delay between the two bands.
- The autocorrelation function is a special case where a function is correlated with itself, which would always show a peak at zero lag.

Convolution is an operation akin to the Cross-correlation, but the function  $g(t)$  in the integrand is inverted to  $g(-t)$  before adding the shift.

- A few important properties of cross-correlation include:

$$[f \star g](\tau) = [g^* \star f^*](\tau)$$

$$[f \star g] \star [f \star g] = [f \star f] \star [g \star g]$$

$$g \star (f \otimes h) = [g \star f] \otimes h$$

$$F[f \star g] = F(f) \cdot F^*(g)$$

- where  $F$  represents the Fourier transform.

We now propose a few definitions for completeness, but without extensive discussions.

## Cross-spectra, phase lag spectra, coherence

- Given two signals  $f(t)$  and  $g(t)$  and their respective FTs,  $F(\omega)$  and  $G(\omega)$ , we define the cross spectrum as:

$$CS(\omega) = F(\omega) \cdot G^*(\omega)$$

- Analogous to the PSD and the autocorrelation, the cross spectrum between two signals is the Fourier transform of their cross-correlation (and vice-versa).
- In its essence, the cross spectrum of two signals at each frequency is a complex number whose argument represents the phase delay between the signals at that frequency.
- The autocorrelation function may be thought of as the second order correlation:

$$c_2(\tau) = \langle f(t)f(t + \tau) \rangle$$

- where the angular brackets denote an ensemble average.
- Bispectrum is an extension of the above concept to triple correlations. The third order correlation function:

$$c_3(\tau_1, \tau_2) = \langle f(t)f(t + \tau_1)f(t + \tau_2) \rangle$$

- that allows one to define the Bispectrum:

$$B(\omega_1, \omega_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c_3(\tau_1, \tau_2) e^{-i(\omega_1 \tau_1 + \omega_2 \tau_2)} d\tau_1 d\tau_2$$

## Power Spectrum statistics

---

- The first thing we need to know is the probability distribution of the noise power.
- We now assume that it is additive and independent of the frequency (i.e. noise is “white”):

$$P_j = P_{j,noise} + P_{j,signal}$$

- The “null hypothesis” is that the periodogram is consistent with pure noise.
- Let’s remind us that If we have  $\mathbf{x}_k \equiv \mathbf{y}_k + \mathbf{z}_k$  and  $\mathbf{b}_j$  and  $\mathbf{c}_j$  are the FT of  $\mathbf{y}_k$  and  $\mathbf{z}_k$ , respectively, we have  $\mathbf{a}_j = \mathbf{b}_j + \mathbf{c}_j$ .
- This does not hold for power spectra unless the signal are uncorrelated random noise:

$$|\mathbf{a}_j|^2 = |\mathbf{b}_j + \mathbf{c}_j|^2 = |\mathbf{b}_j|^2 + |\mathbf{c}_j|^2 + \text{cross terms}$$

- For a wide range of types of noise,  $P_{j,noise}$  follows a  $\chi^2$  distribution with 2 degrees of freedom (but at the Nyquist frequency, where dof is 1).
- For other noise distributions (Poisson, etc.), for large N, due to the central limit theorem the Fourier coefficients  $\mathbf{A}_j$  and  $\mathbf{B}_j$  become still normal.
- This suggests a simple consistency test: compute the standard deviation in each frequency bin and divide by the mean power. Results should be (if the hypotheses hold) close to 1.
- In practice one finds that noise powers are nearly always  $\chi^2$  distributed, not only for Poisson noise, but also for many other types of noise.
- With the Leahy normalization, the probability for  $P_{j,noise}$  to exceed a given threshold is given by:

$$Prob(P_{j,noise} > P_{threshold}) = Q(P_{threshold}|2) \quad (j = 1, N/2 - 1)$$

$$Q(\chi^2|\nu) \equiv \left[ 2^{\nu/2} \Gamma\left(\frac{\nu}{2}\right) \right]^{-1} \int_{\chi^2}^{\infty} t^{\frac{\nu}{2}-1} e^{-\frac{t}{2}} dt$$

- where  $\nu$  is the number of dof and  $\Gamma$  is the gamma function, the generalization of factorial to real and complex numbers [ $\Gamma(n) = (n-1)!$ ]
- For  $\nu = 2$ :

$$Q(\chi^2|2) = \frac{1}{2} \int_{\chi^2}^{\infty} e^{-\frac{t}{2}} dt = e^{-\frac{\chi^2}{2}}$$

### Warning:

Please, pay attention to the adopted periodogram normalization, since this affects the value to plug in the above equation.

- Power spectra are unavoidably very noisy.
  - Standard deviation of noise power is equal to the mean value ( $\sigma_{P_j} = \langle P_j \rangle = 2$ ).
- More interesting, this cannot be improved increasing the number of data points (i.e. the length of the time series). This merely increases the number of powers.
- One can decrease the large variance rebinning the power spectrum and/or dividing the data in multiple segments of equal length. This of course degrade the frequency resolution.
  - If the number of segments is large the power statistics tends to become Normal.
- Let’s define confidence detection level as the power with only  $\epsilon$  probability to be exceeded by noise.

- This holds for a single frequency. If your spectrum consists of  $N_{\text{trial}}$  (independent) frequencies, the confidence detection level decreases to take into account the multiple trials:

$$(1 - \epsilon)^{N_{\text{trial}}} \sim 1 - \epsilon N_{\text{trial}} \quad \text{for } \epsilon \ll 1$$

- In general if noise is not Poissonian or Gaussian noise power spectrum will not be flat anymore.
  - However, often noise powers still follow a  $\chi^2$  distribution with 2 dof, but with a different normalisation (in general depending on  $j$ ).

## The Likelihood for Periodograms

---

- The  $\chi^2$  distribution defines a sampling distribution or likelihood, i.e. the probability distribution of observing a given data set given some underlying (true, unknown) power spectrum.
- If we define a model power  $S_j(\theta)$  at frequency  $\nu_j$ , specified by a set of parameters  $\theta$ , we can then compute the probability of having observed periodogram power  $P_j$  at that same frequency:

$$p(P_j|S_j(\theta)) = \frac{1}{S_j(\theta)} e^{-\frac{P_j}{S_j(\theta)}}$$

- The likelihood (also known as *Whittle likelihood*) for a periodogram over  $N/2$  observed powers  $P_j$  is then defined as the product of individual probabilities for each frequency  $\nu_j$ .
- One generally (and equivalently), defines the logarithm of the likelihood as the sum of logarithm of all probabilities, such that:

$$\log(\mathcal{L}(\theta)) = \sum_{j=1}^{N/2} \log(p(P_j|S_j(\theta))) = - \sum_{j=1}^{N/2} (\log(S_j(\theta)) + \frac{P_j}{S_j(\theta)})$$

- A likelihood for averaged periodograms can be derived from the  $\chi^2_{2LM}/2ML$  sampling distribution for periodograms averaged over  $L$  independent segments and  $M$  independent neighbouring frequencies by:

$$\log(\mathcal{L}_{\text{avg}}(\theta)) = -2ML \sum_{j=1}^{N/2} \left[ \frac{P_j}{S_j(\theta)} + \log(S_j(\theta)) + \left(\frac{1}{ML} - 1\right) \log(P_j) + c(2ML) \right]$$

- where  $c(2ML)$  is a factor independent of  $P_j$  or  $S_j$ , and thus unimportant to the parameter estimation problem considered here.

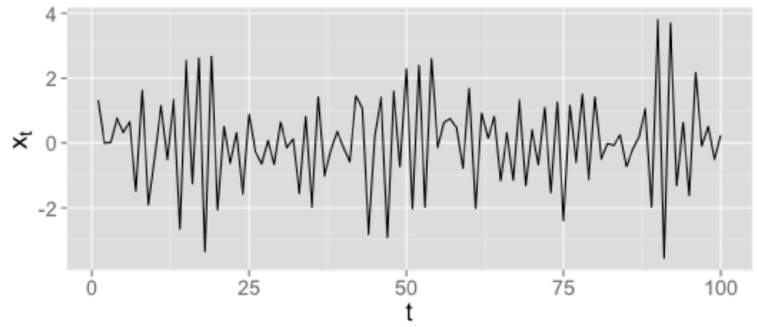
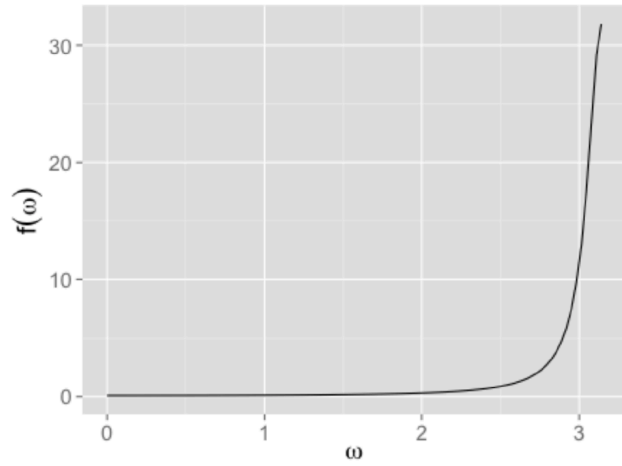
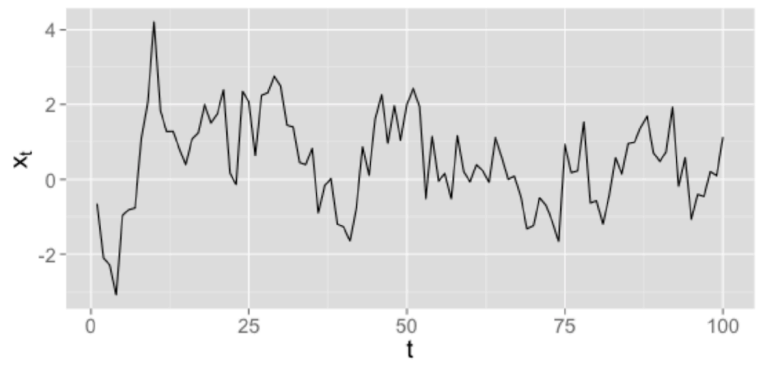
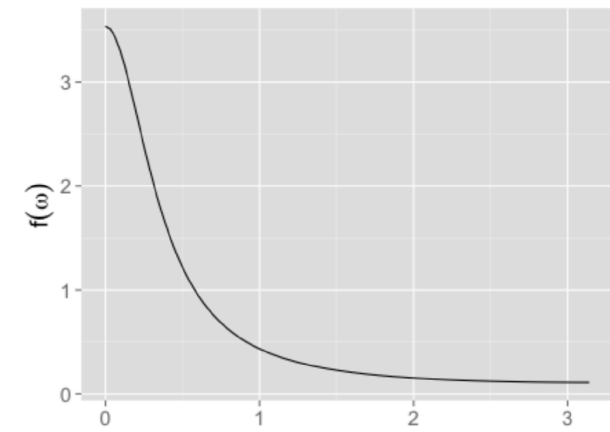
## Noise color

- Fourier analysis (and related techniques) has proven to be very effective in identifying periodic behaviours.
- In astrophysics, however, we often have to deal with phenomena too long for having a reliable coverage (decades or more) and with only approximately cyclical behaviours (quasi-periodicities).
- More important, in order to compute the statistical significance of any possible periodicity, one needs to properly model the noise affecting a time series.

# Coloured Noise



- "Coloured" noise directly affect a time-series shape:



## Final considerations

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- Fourier analysis reveals nothing about the evolution in time, but rather reveals the variance of the signal at different frequencies.
- The classical periodogram is an estimator of the spectral density, i.e. the Fourier transform of the autocovariance function.
- Fourier analysis has restrictive assumptions: an infinitely long data of equally-spaced observations; homoscedastic Gaussian noise with purely periodic signal of sinusoidal shape.
- The classical periodogram is not a good estimator, it is “inconsistent” because the number of parameters grows with the number of datapoints.
- The DFT and its probabilities depends on several strong assumptions that are rarely achieved in real astronomical data: evenly spaced data of infinite duration with a high sampling rate (Nyquist frequency), Gaussian noise, single frequency periodicity with sinusoidal shape and stationary behavior.
- Each of these constraints is often violated in various astronomical problems. Data spacing may be affected by daily/monthly/orbital cycles. Periods may be comparable to the sampling time. Several periods may be present (e.g. helioseismology). Shape may be non-sinusoidal (e.g. elliptical orbits, eclipses, recurrent flares). Periods may not be constant (e.d. QPOs in accretion disks).

## Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Belloni & Bhattacharya \(2022\) - "Basics of Fourier Analysis for High-Energy Astronomy"](#)
- [van der Klis \(1988\) - "Fourier techniques in X-ray timing"](#)

## Further Material

Papers for examining more closely some of the discussed topics.

- [Vaughan \(2010\) - "A Bayesian test for periodic signals in red noise"](#)
- [Barret & Vaughan \(2012\) - "Maximum Likelihood Fitting of X-Ray Power Density Spectra: Application to High-frequency Quasi-periodic Oscillations from the Neutron Star X-Ray Binary 4U1608-522"](#)
- [Covino et al. \(2019\) - "Gamma-ray quasi-periodicities of blazars. A cautious approach"](#)

## Credits

This notebook contains material obtained from <https://www.tutorialspoint.com/power-spectral-density-psd-and-autocorrelation-function#> and from <https://ipython-books.github.io/101-analyzing-the-frequency-components-of-a-signal-with-a-fast-fourier-transform/>.

# Course Flow

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| html     | <a href="#">Lecture about Bayesian statistics</a> | <a href="#">Science case about Sunspot number</a> | <a href="#">Course Summary</a> |

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